

JASPER: Joint Audio and Speech Pre-trained Encoder Representations

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Abstract—Self-supervised learning (SSL) for speech and audio has largely progressed along separate tracks: speech models emphasize time-domain prediction, whereas audio representation learning has focused on time–frequency patterns. This separation creates a compatibility gap, limiting cross-domain generalization. In this work, we introduce JASPER, Joint Audio and Speech Pre-trained Encoder Representations, a framework that augments speech-pretrained models with time–frequency objectives. Specifically, JASPER performs masked prediction of temporal and spectral targets over long audio segments, enabling spectro-temporal representation learning of speech and audio signals. The proposed method consistently outperforms multiple baselines and existing speech/audio encoders on diverse speech, audio, and music tasks, demonstrating the effectiveness of unified spectro-temporal modeling.

Index Terms—Time-frequency masking, Self-supervised learning, Joint speech and audio representations.

I. INTRODUCTION

Self-supervised learning (SSL) has emerged as a dominant paradigm for learning robust representations from unlabeled acoustic data. Existing SSL-based audio models can broadly be categorized into two families based on their input modality: raw waveform-based models and spectrogram-based models. Waveform-based models operate directly on time-domain audio signals, learning hierarchical representations without relying on handcrafted acoustic features. Examples include contrastive predictive frameworks such as wav2vec 2.0 [1], clustering-based masked prediction methods such as HuBERT [2] and WavLM [3]. SSL methods based on the efficient masked autoencoder framework such as Dasheng [4], as well as large-scale weakly-supervised models such as Whisper [5] also provide alternative design choices for audio encoders. For example, the Qwen-audio model uses the Whisper encoder as the audio input processing framework which is fine-tuned for speech and audio tasks [6]. While such waveform-based encoders achieve state-of-the-art performance on speech-centric tasks such as ASR, speaker verification, and speech translation, they fail to generalize to diverse non-speech audio tasks [7].

Spectrogram-based models, in contrast, operate on time–frequency representations such as log-Mel spectrograms. Models such as the Audio Spectrogram Transformer (AST) [8] and Self-Supervised Audio Spectrogram Transformer (SSAST) [9] adopt patch-based tokenization and global self-attention mechanisms to model long-range dependencies across time and

frequency. Spectrogram-based encoders demonstrate strong performance on general audio tasks [10].

Despite their strong performance within their intended domains, the distinct design choices of speech and audio models limit their cross-domain applicability. Frame-level tokenizers used in speech models are less effective at representing complex environmental sounds, whereas patch-based audio models often fall short on speech tasks that depend on precise temporal and sequential information.

While prior works [7] have shown that incorporating temporal learning into spectrogram transformers improves speech performance, the complementary direction remains underexplored. To address these challenges, we propose JASPER - Joint Audio and Speech Pre-trained Encoder Representations. JASPER treats speech and audio pre-training as complementary spectro-temporal objectives over a shared Transformer encoder, jointly predicting temporal targets and spectral patch targets over long audio contexts.

To validate the embedding quality of JASPER and identify the factors contributing to performance gains, we introduce a comprehensive, multi-tiered evaluation framework. First, we apply the frozen representations to traditional intrinsic benchmarking using SUPERB-style [11] downstream linear evaluations, ensuring a direct assessment of acoustic and phonetic properties. Second, we evaluate the extrinsic utility of the embeddings using an encoder-centric evaluation with the X-ARES benchmark [12] and also by deploying JASPER as the unified front-end acoustic encoder for a Large Audio Language Model via the XARES-LLM [13] framework. In these experiments, we highlight the superior performance and generalization of JASPER compared to various other baseline representations.

The main contributions of this work are summarized below:

- We propose JASPER, a unified self-supervised audio encoder that combines waveform-based temporal learning with spectrogram-based spectral learning to improve generalization across speech, environmental audio, and music tasks.
- We introduce a long-context masking strategy over waveform-derived latent representations, where contiguous temporal segments are masked to encourage the encoder to infer missing acoustic content from broader temporal context.
- We propose a spectral prediction objective for waveform encoders by constructing time-aligned spectral

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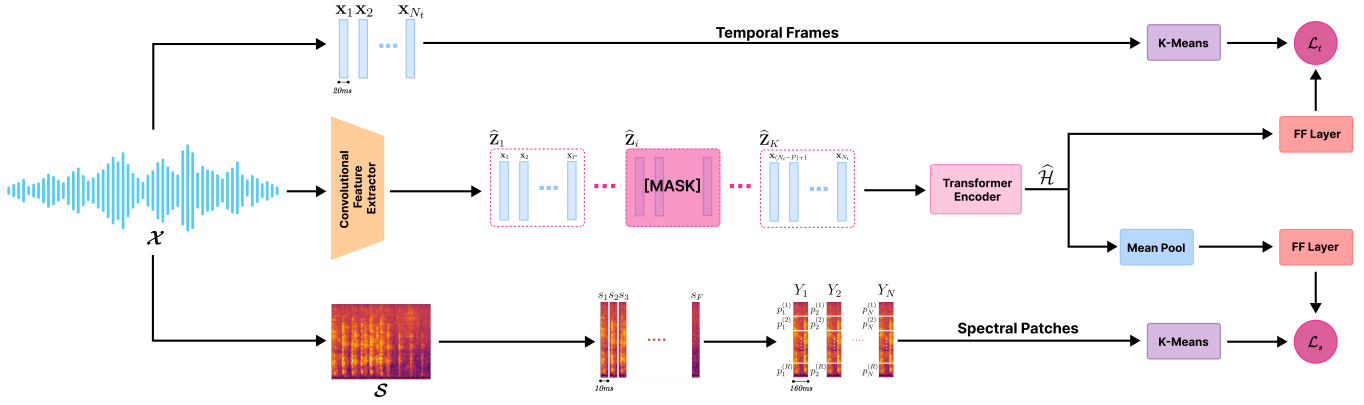


Fig. 1. Block schematic of the proposed JASPER framework for joint spectro-temporal modeling of audio data. ($N_t = 400$, $F = 800$, $N = 50$, $R = 8$).

patch targets from log-Mel representations. The model is trained to predict both temporal pseudo-labels and frequency-localized spectral targets, thereby injecting time-frequency supervision into a raw-waveform encoder without requiring spectrogram input at inference time.

- We establish JASPER as a universal audio encoder through broad speech, audio, music, and audio-LLM evaluations, demonstrating robust representations for both discriminative and generative audio understanding.

II. JASPER ENCODER FRAMEWORK

Figure 1 illustrates the workflow of JASPER consisting of a self-supervised audio encoder designed to learn unified representations for both speech and general audio signals.

A. Input Pre-Processing

Let $\mathcal{X} \in \mathbb{R}^{16000 \times T \times 1}$ denote a raw speech/audio recording of duration T seconds, sampled at 16 kHz. We process $\mathcal{X}$ using a convolutional feature encoder which has an overall stride of 320, resulting in a sequence of latent temporal representations at 50Hz:

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{N_t}], \quad \mathbf{x}_i \in \mathbb{R}^{D_t}, \quad (1)$$

where $N_t = 50T$ denotes the total number of temporal frames and D_t is the dimension of the convolutional encoder.

B. Temporal Segmentation

The raw speech utterance $\mathcal{X}$ is divided into non-overlapping temporal segments, each of duration τ seconds. This segmentation induces a corresponding division of the latent feature sequence $\mathbf{X}$ into non-overlapping windows of P frames, where $P = 50\tau$. Let the windowed latent representation be denoted as

$$\mathbf{Z} = [\mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_K].$$

Here, $\mathbf{Z}_k \in \mathbb{R}^{P \times D_t}$ represents the k -th windowed latent segment, and $K = \frac{T}{\tau}$. $\mathbf{Z}_k$ is constructed by grouping P consecutive latent vectors, i.e.,

$$\mathbf{Z}_k = [\mathbf{x}_{(k-1)P+j}]_{j=1}^P, \quad k = 1, \dots, K.$$

C. Long-Context masking

To enable self-supervised learning, we apply a masked modeling strategy to the windowed latent representations $\mathbf{Z}$. Each latent window is independently masked with probability p . Furthermore, to encourage the model to capture longer temporal dependencies, if a window is selected for masking, its subsequent window is also masked with a fixed probability p' .

Let the resulting masked latent sequence be denoted as:

$$\hat{\mathbf{Z}} = [\hat{\mathbf{Z}}_1, \hat{\mathbf{Z}}_2, \dots, \hat{\mathbf{Z}}_K].$$

The masking operation is defined as:

$$\hat{\mathbf{Z}}_k = \begin{cases} [\text{MASK}], & \text{if location } k \text{ is selected for masking,} \\ \mathbf{Z}_k, & \text{otherwise,} \end{cases}$$

where $[\text{MASK}] \in \mathbb{R}^{P \times D_t}$ denotes a learnable mask embedding (or a zero mask of the same dimensionality). In our experiments, we use $p = 0.6$ and $p' = 0.2$.

D. Transformer Encoding of Temporal Frames

To retain temporal order information, a convolutional positional embedding is added to the frame sequence within each segment. The resulting sequence is then passed to a transformer encoder, which produces contextualized temporal representations, each of dimension D' :

$$\hat{\mathbf{H}}_k = [\hat{\mathbf{h}}_k^{(1)}, \hat{\mathbf{h}}_k^{(2)}, \dots, \hat{\mathbf{h}}_k^{(P)}], \quad k = 1, \dots, K \quad \hat{\mathbf{h}}_k^{(i)} \in \mathbb{R}^{D'}.$$

E. Temporal Targets and Loss Function

For each $\mathbf{x}_i$, a discrete pseudo-label $\mathbf{C}_{t,k}^{(i)}$ is obtained by clustering the frame-level features $\mathbf{x}_i$ using K-means clustering. We define a temporal loss function $\mathcal{L}_t$ that encourages the transformer encoder of the JASPER model to predict these pseudo-labels. The loss is computed only on the masked frames.

Specifically, a K-means quantizer $\mathcal{Q}$ is applied to frame-level embeddings to generate cluster assignments:

$$\mathbf{C}_{t,k}^{(i)} = \mathcal{Q}(\mathbf{x}_i), \quad \mathbf{C}_{t,k}^{(i)} \in \{1, 2, \dots, V_t\}$$

$$k = 1, \dots, K \quad i = 1, \dots, P$$

where $\mathbf{C}_{t,k}^{(i)}$ denotes the cluster label corresponding to the i -th frame in the k -th window and V_t denotes the number of clusters. Given the masked sequence $(\hat{\mathbf{Z}})$ as input, the transformer encoder produces contextualized representations for all temporal frames. Let $\hat{\mathbf{h}}_k^{(i)} \in \mathbb{R}^{D'}$ denote the output corresponding to the i -th frame of the k -th window. Each frame representation is projected using a learnable feed-forward layer, with parameters $\mathbf{W}_t \in \mathbb{R}^{V_t \times D'}$, followed by a softmax operation:

$$\hat{\mathbf{p}}_k^{(i)} = \text{softmax}(\mathbf{W}_t \hat{\mathbf{h}}_k^{(i)}).$$

The training objective is computed only over masked frame indices. The temporal loss is defined as:

$$\mathcal{L}_t = \frac{1}{|\mathcal{M}|} \sum_{k \in \mathcal{M}} \sum_{i=1}^P \text{C.E.}(\hat{\mathbf{p}}_k^{(i)}, \mathbf{C}_{t,k}^{(i)}) \quad (2)$$

where $\mathcal{M}$ denotes the set of masked window indices, and $\text{C.E.}(\cdot, \cdot)$ denotes the standard cross-entropy loss.

F. Spectral Targets and Loss

Let the log-Mel spectrogram of waveform $\mathcal{X}$ be

$$\mathcal{S} = [\mathbf{s}_1, \dots, \mathbf{s}_F],$$

where $\mathbf{s}_i \in \mathbb{R}^D$, $D = 128$, extracted using 128-D Mel-filterbank features (25 ms window, 10 ms hop). The spectrogram is divided into non-overlapping segments of Q consecutive spectral frames corresponding to the temporal duration τ seconds:

$$\mathcal{Y} = [\mathbf{Y}_1, \dots, \mathbf{Y}_N], \quad \mathbf{Y}_n \in \mathbb{R}^{Q \times D}, \quad N = \frac{F}{Q}.$$

Each spectrogram segment $\mathbf{Y}_n$ captures the spectral structure of the windowed signal of τ seconds duration. Each $\mathbf{Y}_n$ is further split vertically into R non-overlapping $Q \times Q$ square patches:

$$\mathbf{P}_n = \begin{bmatrix} \mathbf{p}_n^{(1)} & \mathbf{p}_n^{(2)} & \dots & \mathbf{p}_n^{(R)} \end{bmatrix}^\top, \quad \mathbf{p}_n^{(r)} \in \mathbb{R}^{Q \times Q}, \quad R = \frac{D}{Q}.$$

Spectral targets are obtained by flattening each patch and quantizing via a K-means quantizer Q_s :

$$C_{s,k}^{(r)} = Q_s(\text{vec}(\mathbf{p}_n^{(r)}))$$

$$C_{s,k}^{(r)} \in \{1, \dots, V_s\}, n = 1, \dots, N, r = 1, \dots, R.$$

Here, V_s denotes the number of spectral K-means clusters. Consider a temporal segment-level representation for the transformer embeddings via mean pooling $\bar{h}_k = \frac{1}{P} \sum_{i=1}^P \hat{h}_k^{(i)}$. We design R independent feed-forward heads $\{W_s^{(r)}\}_{r=1}^R$ to predict spectral cluster labels as:

$$p_{s,k}^{(r)} = \text{softmax}(W_s^{(r)} \bar{h}_k).$$

The spectral loss is computed over masked windows:

$$L_s = \frac{1}{|M|R} \sum_{k \in M} \sum_{r=1}^R \text{C.E.}(p_{s,k}^{(r)}, C_{s,k}^{(r)}).$$

where C.E. is the standard cross-entropy loss and $|M|$ is the number of masked frames.

G. Combined Training Objective

The overall training objective of the model is the sum of spectral loss $\mathcal{L}_s$ and temporal loss $\mathcal{L}_t$:

$$\mathcal{L}_{\text{total}} = \lambda \mathcal{L}_t + (1 - \lambda) \mathcal{L}_s \quad (3)$$

III. IMPLEMENTATION DETAILS

A. Model Architecture and Targets

Each input audio signal is truncated or zero-padded to 8 s and the audio segment duration $\tau = 0.16$ s is chosen to match syllable-length units. The raw waveform is processed by a convolutional feature encoder with $D_t = 512$ and passed through a 12-layer transformer encoder with 12 attention heads, and $D' = 768$. For self-supervised target construction, we use $V_t = 500$ and $V_s = 500$.

For the spectral processing, we have $F = 800$ and $Q = 16$. This gives $N = 50$ non-overlapping windows. The number of vertical patches in each spectral segment is $R = 8$.

B. Pre-training Setup and Optimization

JASPER is pretrained under two data regimes: 200-hour and 2000-hour settings, each comprising equal amounts of speech and audio from LibriSpeech [14] and AudioSet [15]. The encoder is first initialized using temporal-only targets, followed by joint predictive pre-training with combined temporal and spectral objectives for 1M steps. Training uses the AdamW optimizer with weight decay of 0.05, a constant learning rate of 2×10^{-5} , and coefficients $\beta = [0.9, 0.98]$.

IV. EVALUATION SETUP

We evaluate the proposed audio encoder through a comprehensive suite of downstream tasks and evaluation protocols. First, we assess the quality of the learned representations via a linear evaluation framework inspired by the SUPERB benchmark [11]. Under this protocol, the pre-trained encoder remains frozen and serves solely as a feature extractor. Representations across all transformer layers are aggregated using a learned weighted sum, and only the aggregation weights and a lightweight, task-specific prediction head are optimized during evaluation. This approach measures representation quality independent of task-specific fine-tuning. We evaluate the model on diverse downstream tasks spanning speech, environmental sound, and music domains, covering both clip-level and frame-level prediction settings. Specifically, the model is evaluated on the ESC-50 [23], UrbanSound8K (US8k) [24], GTZAN [25], FMA Small [26], IEMOCAP [27], VoxCeleb1 (VoxC1) [28], Speech Commands V2 [29], CREMA-D [30], LibriCount [31], TIMIT [32] and LibriSpeech [14] datasets. Performance is reported using standard task-specific metrics, including classification accuracy, Phoneme Error Rate (PER), Word Error Rate (WER), and Character Error Rate (CER).

We further benchmark the effectiveness of the learned representations using two complementary evaluation settings: (i) encoder-centric evaluation using the X-ARES benchmark

TABLE I

DOWNSTREAM PERFORMANCE COMPARISON UNDER 200-HOUR AND 2000-HOUR PRE-TRAINING SETUPS. ALL MODELS ARE TRAINED USING THE SAME DATA AND NUMBER OF PRE-TRAINING STEPS. RESULTS ARE REPORTED AS CLASSIFICATION ACCURACY (%).

Model	Audio Tasks				Speech Tasks				
	ESC-50	US8k	GTZAN	FMA_SMALL	IEMOCAP	VoxC1	SPCV2	CREMA-D	LibriC.
200 hour setup									
SSAST [9]+MLM	77.65	78.46	80.08	53.75	60.49	32.19	78.89	54.60	68.77
HuBERT [2]	66.10	70.05	79.88	48.12	59.91	28.83	84.82	60.35	68.18
JASPER	83.55	78.84	89.69	52.5	65.02	53.93	92.80	70.01	66.91
2000 hour setup									
SSAST [9]+MLM	85.50	84.27	91.29	49.11	62.05	47.95	82.03	70.49	70.73
HuBERT [2]	80.55	78.42	91.89	49.25	66.27	64.56	95.55	68.46	67.55
JASPER	88.05	83.66	91.99	53.75	68.76	91.18	94.78	76.89	72.13

US8k: UrbanSound8K; FMA: Free Music Archive; VoxC1: VoxCeleb1; SPCV2: Speech Commands V2; LibriC.: LibriCount

TABLE II

COMPARISON OF THE PROPOSED JASPER WITH BASELINE MODELS UNDER THE 2000-HOUR PRE-TRAINING SETUP. ALL RESULTS ARE REPORTED AS CLASSIFICATION ACCURACY (%).

Model	#Params	Data (hrs)	Audio				Speech			Avg.
			ESC-50	US8k	GTZAN	FMA_SMALL	IEMOCAP	VoxC1	SPCV2	
HuBERT [2]	95M	1k	77.45	77.61	74.00	42.25	68.30	86.68	96.10	74.63
SSAST [9]	89M	5.8k	37.85	54.87	77.10	49.38	55.21	15.62	37.87	46.84
BEATs [10]	91M	5.8k	92.45	83.21	89.01	58.13	67.61	56.84	92.30	77.08
EnCodecMAE [16]	86.6M	11k	88.25	85.42	87.10	56.20	67.80	71.65	97.30	79.10
USAD [17]	94M	161k	87.45	84.09	87.05	54.63	70.61	69.85	96.65	78.62
ATST [18]	86M	5.3k	92.90	87.60	90.01	55.25	67.79	68.29	92.53	79.20
ULTRAS [7]	89M	2k	91.15	86.07	92.59	55.62	67.78	73.55	95.10	80.27
JASPER	95M	2k	88.05	83.66	91.99	53.75	68.76	91.18	94.78	81.74

US8k: UrbanSound8K; FMA: Free Music Archive; VoxC1: VoxCeleb1; SPCV2: Speech Commands V2.

TABLE III

COMPARISON OF PHONEME RECOGNITION (PR) AND AUTOMATIC SPEECH RECOGNITION (ASR) PERFORMANCE. IN THE CASE OF HUBERT (OURS), THE PRE-TRAINING DATA IS A MIXTURE OF SPEECH AND AUDIO (2000H) SIMILAR TO THE JASPER, WHILE THE OFFICIAL IMPLEMENTATION USES ONLY 1000H OF LIBRISPEECH DATA .

Model	TIMIT	LibriSpeech		
	PER (%)	PER (%)	CER (%)	WER (%)
HuBERT (Official)	11.9	7.1	2.4	8.5
HuBERT (Ours)	13.0	7.8	3.7	12.4
JASPER	11.9	6.9	3.3	10.9

PER: Phoneme Error Rate; CER: Character Error Rate; WER: Word Error Rate

[12], and (ii) audio-LLM suitability evaluation using the XARES-LLM framework [13]. The X-ARES encoder-centric evaluation employs two distinct methodologies: Multi-Layer Perceptron (MLP) linear probing and k -Nearest Neighbors (k -NN) non-parametric evaluation [12]. For the MLP evaluation, the encoder and task-specific components remain frozen, and only a single linear layer is trained to assess representation quality. Conversely, the k -NN evaluation is entirely non-parametric, classifying extracted embeddings based solely on their proximity within the feature space.

The audio-LLM suitability evaluation is conducted by routing the X-ARES pipeline through a Large Audio Language Model (LALM) backbone [13]. In this setting, the frozen encoder embeddings provide the acoustic representations that

TABLE IV

WEIGHTED AVERAGE PERFORMANCE ACROSS TASKS UNDER MLP AND k -NN EVALUATION SETTINGS UNDER XARES EVALUATION BENCHMARK (HIGHER VALUES INDICATE BETTER PERFORMANCE)

Model	MLP	k-NN
ATST-Clip [18]	0.481	0.433
ATST-Frame [18]	0.622	0.437
BEATs [10]	0.451	0.361
BYOL-A [19]	0.487	0.239
BYOL-S [20]	0.391	0.259
CED [21]	0.496	0.548
Dasheng [4]	0.696	0.499
data2vec [22]	0.565	0.379
HuBERT [2]	0.602	0.207
MSM-MAE [19]	0.444	0.437
w2v2 [1]	0.384	0.254
WavLM [3]	0.525	0.437
Whisper [5]	0.646	0.301
JASPER	0.702	0.476

Results of standard models are tabulated from the official X-ARES benchmark [12]

are used to guide an autoregressive text generator across open-ended classification and captioning tasks. The language modeling core of the XARES-LLM pipeline is driven by SmolLM2-135M [33], a highly optimized, lightweight language model trained extensively on curated textual corpora.

TABLE V
COMPARISON OF JASPER ENCODER WITH OTHER BASELINE MODELS - TRACK A (HIGHER VALUE IS BETTER FOR ALL TASKS)

Model	Params	Hrs	Speech Tasks								Audio Tasks				Music Tasks			Avg.
			SC (Acc)	LibriC. (Acc)	VoxL33 (Acc)	VoxC1 (Acc)	ASV15 (Acc)	FSC (Acc)	VocalS. (Acc)	CREMA-D (Acc)	ESC-50 (Acc)	FSD50k (mAP)	US8k (Acc)	FSD18 (mAP)	GTZAN (Acc)	NSynth (Acc)	FMA (Acc)	
HuBERT [2]	94.7M	1k	0.65	0.41	0.72	0.72	0.90	0.99	0.88	0.54	0.39	0.05	0.54	0.32	0.69	0.57	0.52	0.59
WavLM [3]	94.7M	60k	0.67	0.31	0.74	0.68	0.90	0.99	0.86	0.57	0.34	0.04	0.48	0.25	0.62	0.52	0.46	0.56
Whisper [5]	74.0M	680k	0.69	0.41	0.83	0.76	0.94	0.82	0.87	0.52	0.63	0.09	0.74	0.55	0.70	0.64	0.58	0.65
Dasheng [4]	86.4M	272k	0.65	0.39	0.31	0.97	0.94	0.98	0.85	0.62	0.75	0.06	0.83	0.41	0.32	0.67	0.43	0.61
w2v2 [1]	94.4M	1k	0.15	0.47	0.23	0.60	0.81	0.33	0.86	0.42	0.32	0.04	0.43	0.25	0.65	0.48	0.47	0.43
ULTRAS [7]	89M	2k	0.52	0.46	0.28	0.84	0.95	0.86	0.91	0.52	0.60	0.08	0.79	0.5	0.78	0.62	0.56	0.62
JASPER	95.0M	2k	0.69	0.45	0.79	0.74	0.95	0.99	0.90	0.55	0.56	0.07	0.76	0.43	0.80	0.63	0.57	0.66

SC: Speech Commands; LibriC.: LibriCount; VoxL33: VoxLingual 33; VoxC1: VoxCeleb1; ASV15: ASVspoof 2015; FSC: Fluent Speech Commands; VocalS.: VocalSound; US8k: UrbanSound8K; FMA: Free Music Archive; Acc: Accuracy; mAP: Mean Average Precision.

TABLE VI
COMPARISON OF JASPER ENCODER WITH OTHER BASELINE MODELS - TRACK B (HIGHER VALUE IS BETTER FOR ALL TASKS)

Model	Params	Hrs	AISHELL-1 (iWER)	Clotho (FENSE)	LibriSpeech (iWER)	MECAT (DATE)	Song-Describer (FENSE)	Avg.
HuBERT [2]	94.7M	1k	0.14	0.27	0.78	0.58	0.42	0.44
WavLM [3]	94.7M	60k	0.19	0.21	0.84	0.58	0.45	0.45
Whisper [5]	74.0M	680k	0.36	0.36	0.38	0.62	0.45	0.40
Dasheng [4]	86.4M	272k	0.02	0.21	0.10	0.60	0.41	0.27
w2v2 [1]	94.4M	1k	0.0	0.26	0.0	0.54	0.43	0.25
JASPER	95.0M	2k	0.27	0.35	0.81	0.61	0.46	0.50

iWER: 1 - WER ; FENSE: Fluency-Enhanced Sentence-bert Evaluation; DATE: Discriminative-Enhanced Audio Text Evaluation

For the XARES-LLM evaluation, downstream performance is mapped across two distinct evaluation pathways:

A. Track A - Classification Tasks:

This track evaluates encoder representations on utterance-level classification tasks in domains including speech, environmental sounds, and music. Performance is measured using accuracy (Acc) and mean average precision (mAP).

B. Track B - Understanding Tasks:

This track assesses higher-level semantic understanding using generative and sequence prediction tasks like automatic speech recognition, audio captioning, and music captioning. Output quality is assessed using inverted Word Error Rate (iWER), Fluency-Enhanced Sentence-bert Evaluation (FENSE), and Discriminative-Enhanced Audio Text Evaluation (DATE).

V. RESULTS AND DISCUSSION

A. Linear Evaluation

1) *Clip-level Classification*: Table I reports the classification accuracy of different self-supervised audio models evaluated on nine diverse downstream tasks under the two pre-training data regimes: 200 hours and 2000 hours. All models are pretrained on the same dataset and follow the open-source pre-training recipes provided in their respective repositories. We compare HuBERT [2] which uses two-stage temporal prediction-based pre-training, SSAST [9] trained with a masked-language modeling (MLM) objective instead of the reconstruction and location-identification losses proposed

in the original framework [9], and JASPER. The original loss framework for SSAST has been changed for ensuring a fair comparison with the other models. Results show that the JASPER framework consistently outperforms both spectral and temporal baselines across the 200-hour and 2000-hour pre-training regimes on the majority of audio and speech tasks. Table II compares JASPER with existing state-of-the-art audio self-supervised models. Although JASPER is pretrained with only 2000 hours of data, which is much lower than several high-compute baselines, it remains highly competitive across both audio and speech tasks and achieves the best overall average accuracy of 81.74%. These results highlight that the proposed joint spectro-temporal masking strategy enables JASPER to learn efficient and transferable representations with substantially less pre-training data.

2) *Frame-level Sequence Prediction*: To evaluate whether the encoder preserves fine-grained temporal information, we test frozen representations on frame-level sequence-to-sequence tasks using a lightweight linear CTC head. Table III reports phoneme recognition (PR) on TIMIT and LibriSpeech under linear evaluation, and ASR on LibriSpeech-100 using a BiLSTM decoder, with CER and WER as metrics. The results show that JASPER retains detailed speech-level information despite adding spectral prediction and audio-domain pre-training objectives.

B. Encoder-Centric Benchmark Evaluation (X-ARES)

As presented in Table IV, JASPER achieves best performance under the parametric MLP probing setting. In the non-parametric k -NN setting, JASPER retains a highly robust

TABLE VII

RESULTS ON TRACK A TASKS UNDER THE XARES-LLM FRAMEWORK FOR MODELS PRE-TRAINED ON 200 HOUR AND 2000 HOUR SETUP. MODELS ARE EVALUATED USING IDENTICAL DOWNSTREAM FINE-TUNING AND EVALUATION PROTOCOLS. (HIGHER VALUE IS BETTER FOR ALL TASKS)

200 hour setup																
Model	Speech Tasks								Audio Tasks				Music Tasks			Avg.
	SC (Acc)	LibriC. (Acc)	VoxL33 (Acc)	VoxC1 (Acc)	ASV15 (Acc)	FSC (Acc)	VocalS. (Acc)	CREMA-D (Acc)	ESC-50 (Acc)	FSD50k (mAP)	US8k (Acc)	FSD18 (mAP)	GTZAN (Acc)	NSynth (Acc)	FMA (Acc)	
HuBERT [2]	0.43	0.44	0.21	0.68	0.89	0.63	0.87	0.50	0.52	0.05	0.65	0.38	0.75	0.61	0.52	0.54
JASPER	0.43	0.45	0.26	0.85	0.96	0.72	0.88	0.52	0.53	0.06	0.72	0.42	0.70	0.64	0.55	0.58
2000 hour setup																
HuBERT [2]	0.70	0.43	0.78	0.51	0.96	0.99	0.89	0.52	0.51	0.07	0.72	0.39	0.82	0.63	0.59	0.63
JASPER	0.69	0.45	0.79	0.74	0.95	0.99	0.90	0.55	0.56	0.07	0.76	0.43	0.80	0.63	0.57	0.66

SC: Speech Commands; LibriC.: LibriCount; VoxL33: VoxLingual 33; VoxC1: VoxCeleb1; ASV15: ASVspoof 2015; FSC: Fluent Speech Commands; VocalS.: VocalSound; US8k: UrbanSound8K; FMA: Free Music Archive; Acc: Accuracy; mAP: Mean Average Precision.

score, outperforming traditional speech baselines like HuBERT [2]. The competitive performance across the parametric and non-parametric evaluations demonstrates that our joint spectro-temporal objective yields a well-structured representation space that naturally clusters by semantic similarity without relying on classifier training.

C. XARES-LLM Evaluation

Tables V and VI present the XARES-LLM results for tasks routed through the LALM backbone. In Track A, JASPER achieves the best average score of 0.66, outperforming large-scale baselines such as Whisper [5] despite using far less pre-training data (2k vs. 680k hours). While Dasheng [4] and Whisper show task-specific strengths, JASPER provides a more balanced cross-domain performance profile. In Track B, JASPER again leads with an average score of 0.50, surpassing WavLM [3] and HuBERT [2]. These results suggest that JASPER’s joint spectro-temporal representations provide strong and transferable features for speech and audio understanding.

VI. ABLATION STUDIES

A. Scaling Improvements

We study the effect of pre-training scale by training JASPER with 200 hours and 2000 hours of unlabeled speech-audio data under the same architecture and objective. Increasing the pre-training data consistently improves downstream performance across speech, environmental audio, and music tasks as shown in Table I. This confirms that the joint spectro-temporal objective is scalable. We further analyze scaling under the XARES-LLM protocol in Table VII, where increasing JASPER pre-training from 200 hours to 2000 hours leads to consistent gains when the encoder is used as the acoustic front-end of an audio-language model.

B. Effect of Loss Weight λ

To investigate the sensitivity of JASPER to the loss weight λ in Eq. (3), we conduct an ablation study over different values of λ . As shown in Table VIII, the best performance is obtained at $\lambda = 0.01$, where the spectral component receives a stronger weight while retaining a small contribution from the temporal objective. This trend is consistent with our central hypothesis: adding spectral prediction to a waveform encoder provides

TABLE VIII

PERFORMANCE OF JASPER FOR DIFFERENT λ VALUES IN THE 200 HR SETUP. RESULTS ARE REPORTED AS CLASSIFICATION ACCURACY (%).

λ	Audio Tasks			Speech Tasks		
	ESC-50	US8k	GTZAN	IEMOCAP	VoxC1	SPCV2
0.01	83.55	78.84	89.69	65.02	53.93	92.80
0.25	82.10	78.05	88.99	63.85	53.12	91.95
0.50	80.30	77.83	88.89	63.46	51.23	91.23
0.75	79.70	76.65	88.99	63.19	51.67	91.14

US8k: UrbanSound8K; VoxC1: VoxCeleb1; SPCV2: Speech Commands V2.

useful frequency-domain supervision, leading to stronger performance on audio tasks. At the same time, the temporal loss preserves speech-relevant sequential information, enabling competitive performance on speech tasks.

VII. CONCLUSION

We introduced JASPER, a unified self-supervised framework for joint audio-speech representation learning. By integrating complementary time-frequency predictive objectives with temporal modeling, JASPER unifies speech and generic audio characteristics within a single raw-waveform encoder. Extensive benchmarking across a diverse suite of SUPERB-style diagnostic tasks and the XARES-LLM pipeline demonstrates that the continuous representations extracted by JASPER consistently outperform other established baselines across both frame-level and utterance-level tasks. These results highlight the benefits of unified spectro-temporal modeling for capturing the complex acoustic and semantic information required for large audio language models. Furthermore, all the improvements in the proposed framework are achieved with significantly smaller amounts of training data compared to several other baseline models.

VIII. AI GENERATED CONTENT DISCLOSURE

All the scientific content, methodology, analysis, results and conclusions were developed by the authors who take full responsibility of the work. Generative AI tools were used only for minor grammatical corrections to improve the readability of the manuscript.

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